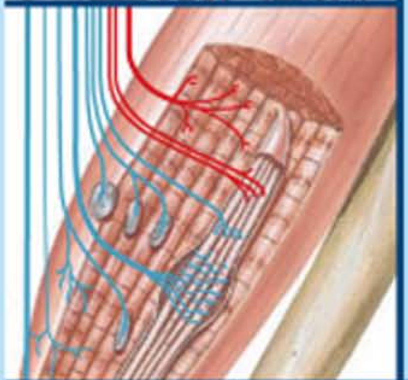
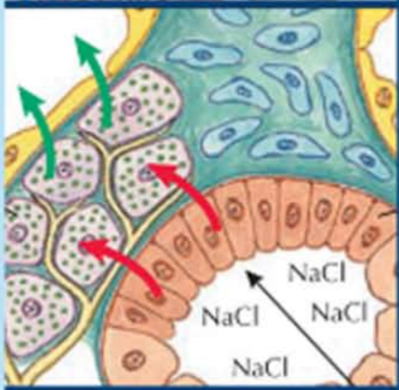
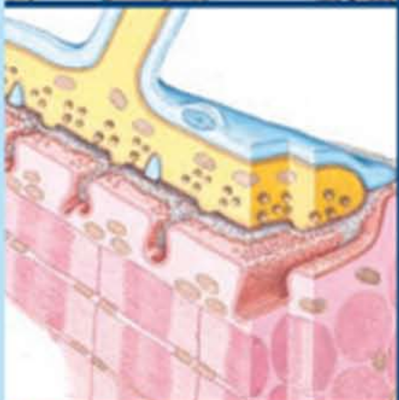
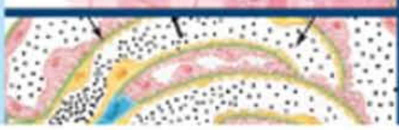


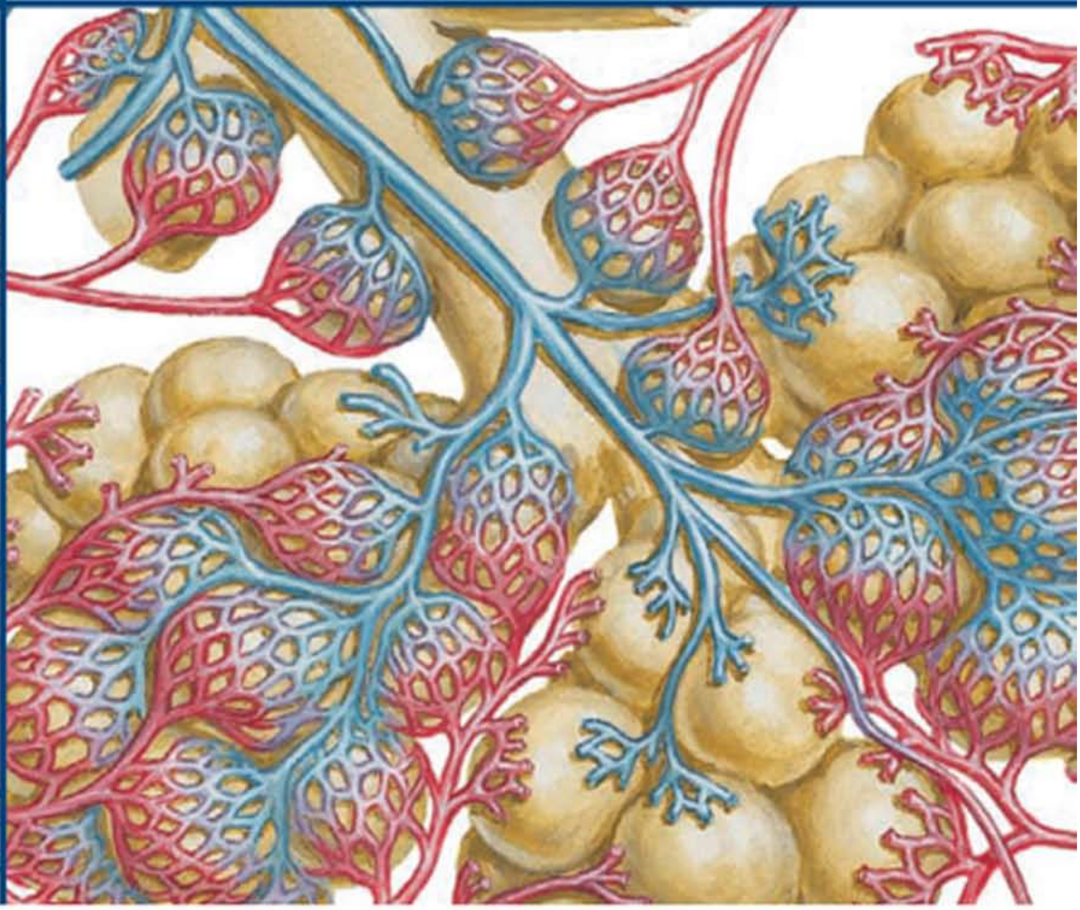






NETTER'S ESSENTIAL PHYSIOLOGY

SUSAN MULRONEY • ADAM MYERS



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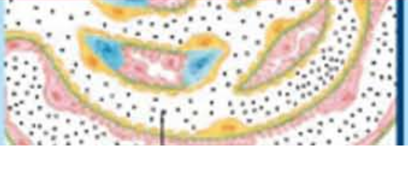

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NETTER'S ESSENTIAL PHYSIOLOGY

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We dedicate this book to our families, for their love and support and for their patience during the preparation of this book. We dedicate it also to the students of Georgetown University, who are exceptional in their character and their love of learning.

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PREFACE

Human physiology is the study of the functions of our bodies at all levels: whole organism, systems, organs, tissues, cells, and physical and chemical processes. Physiology is a complex science, incorporating concepts and principles from biology, chemistry, biochemistry, and physics; and often, a true appreciation of physiological concepts requires multiple learning modalities, beyond standard texts or lectures. This book, *Netter's Essential Physiology,* has been prepared with this in mind. Its generous illustrations and concise, bulleted, and highlighted text are designed to draw the student in, to focus the student's efforts on understanding the essential aspects of difficult concepts. It is intended not to be a detailed reference book, but rather a guide to learning the essentials of the field, in conjunction with classroom work and other texts when necessary.

This book is organized in the classical order in which subdisciplines of physiology are taught. Beginning with fluid compartments, transport mechanisms, and cell physiology, it progresses through neurophysiology, cardiovascular physiology, the respiratory system, renal physiology, the gastrointestinal system, and endocrinology. It is ideal for the visual learner. Each section is thoroughly illustrated with the great drawings of the late Frank Netter as well as the more recent, beautiful work of Carlos Machado, John Craig, and James Perkins.

Recognizing that physiology, cell biology, and anatomy go hand in hand in the modern, integrated curriculum of many institutions, we have included more than the usual number of illustrations relevant to anatomy and histology. By reading the text, studying the illustrations, and taking advantage of the review questions, the student will become familiar with the important concepts in each subdiscipline and gain the essential knowledge required in medical, dental, upper level undergraduate, or nursing courses in human physiology.

Too many textbooks, although very useful reference works, go for the most part unread by students. It is our hope that students will find this book enriching and stimulating and that it will inspire them to thoroughly learn this fascinating field.

Susan E. Mulroney, PhD
Adam K. Myers, PhD

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Acknowledgments

The preparation of this textbook has benefited from the efforts of numerous colleagues and students who reviewed various sections of the work and offered valuable criticisms and suggestions. We especially wish to thank Charles Read, Henry Prange, Stefano Vicini, Jagmeet Kanwal, Peter Kot, Edward Inscho, Jennifer Rogers, Adam Mitchell, Milica Simpson, Lawrence Bellmore, and Joseph Garman for their critical reviews. In addition, we express our appreciation to Adriane Fugh-Berman, whose insights and advice helped us avert many potential nightmares; Amy Richards, for her constant good humor and willingness to help; and all our colleagues and coworkers for their friendship, collegiality, and encouragement during this project.

Our special thanks go to the dedicated team at Elsevier, particularly Marybeth Thiel and Elyse O'Grady. We also acknowledge Jim Perkins for his talented work on the new illustrations in this volume, which gracefully complement the original drawings of the master illustrator, Frank Netter.

Finally, we acknowledge the role of our students in this project, for their encouragement and for their enthusiasm in learning, which is the greatest inspiration for our work.

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About the Authors

SUSAN E. MULRONEY, PhD, is an award-winning teacher and researcher at Georgetown University Medical Center, where she is Professor of Physiology and Biophysics and Director of the highly acclaimed Physiology Special Master's Program. Dr. Mulroney directed the Medical Human Physiology course for the first-year medical students at the School of Medicine and now is Director of the Medical Gastrointestinal module in the new systems-based medical curriculum. She also directs the Medical Physiology course for the Georgetown Summer Medical Institute. Dr. Mulroney lectures to medical and graduate students in multiple areas of human physiology, including renal, gastrointestinal, and endocrine physiology, and is recognized for her expertise in curricular innovation in medical education. Dr. Mulroney is a well-known researcher in renal and endocrine physiology, has published extensively in these areas, and was director of the Physiology PhD program for 12 years. She is also coeditor of *RNA Binding Proteins: New Concepts in Gene Regulation.*

ADAM K. MYERS, PhD, is Professor of Physiology and Biophysics and Associate Dean for Graduate Education at Georgetown University Medical Center. Dr. Myers was director of the Special Master's Program at Georgetown for 12 years and developed and co-directed the M.S. program in Complementary and Alternative Medicine. He also founded and directs the Georgetown Summer Medical Institute. He has won numerous teaching awards from the students and faculty of Georgetown University School of Medicine, where he teaches extensively in several medical and graduate courses in various areas of human physiology. Dr. Myers is recognized for his extensive experience in educational program development and administration. His ongoing research in platelet and vascular biology has resulted in numerous publications. He is also author of the textbook *Crash Course: Respiratory System* and coeditor of *Alcohol and Heart Disease.*

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About the Artists

FRANK H. NETTER, MD, was born in 1906 in New York City. He studied art at the Art Student's League and the National Academy of Design before entering medical school at New York University, where he received his MD degree in 1931. During his student years, Dr. Netter's notebook sketches attracted the attention of the medical faculty and other physicians, allowing him to augment his income by illustrating articles and textbooks. He continued illustrating as a sideline after establishing a surgical practice in 1933, but he ultimately opted to give up his practice in favor of a full-time commitment to art. After service in the United States Army during World War II, Dr. Netter began his long collaboration with the CIBA Pharmaceutical Company (now Novartis Pharmaceuticals). This 45-year partnership resulted in the production of the extraordinary collection of medical art so familiar to physicians and other medical professionals worldwide.

In 2005, Elsevier Inc. purchased the Netter Collection and all publications from Icon Learning Systems. There are now more than 50 publications featuring the art of Dr. Netter available through Elsevier Inc. (in the United States: www.us.elsevierhealth.com/Netter and outside the United States: www.elsevierhealth.com).

Dr. Netter's works are among the finest examples of the use of illustration in the teaching of medical concepts. The 13-book *Netter Collection of Medical Illustrations,* which includes the greater part of the more than 20,000 paintings created by Dr. Netter, became and remains one of the most famous medical works ever published. *The Netter Atlas of Human Anatomy,* first published in 1989, presents the anatomical paintings from the Netter Collection. Now translated into 16 languages, it is the anatomy atlas of choice among medical and health professions students the world over.

The Netter illustrations are appreciated not only for their aesthetic qualities but, more importantly, for their intellectual content. As Dr. Netter wrote in 1949, ". . . clarification of a subject is the aim and goal of illustration. No matter how beautifully painted, how delicately and subtly rendered a subject may be, it is of little value as a *medical illustration* if it does not serve to make clear some medical point." Dr. Netter's planning, conception, point of view, and approach are what informs his paintings and what makes them so intellectually valuable.

Frank H. Netter, MD, physician and artist, died in 1991.

Learn more about the physician-artist whose work has inspired the Netter Reference collection: http://www.netterimages.com/artist/netter.htm.

CARLOS A.G. MACHADO, MD, was chosen by Novartis to be Dr. Netter's successor. He continues to be the main artist who contributes to the Netter collection of medical illustrations.

Self-taught in medical illustration, cardiologist Carlos Machado has contributed meticulous updates to some of Dr. Netter's original plates and has created many paintings of his own in the style of Netter as an extension of the Netter collection. Dr. Machado's photorealistic expertise and his keen insight into the physician/patient relationship informs his vivid and unforgettable visual style. His dedication to researching each topic and subject he paints places him among the premier medical illustrators at work today.

Learn more about his background and see more of his art at http://www.netterimages.com/artist/machado.htm.

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Section 5: **Renal Physiology**

Section 6: **Gastrointestinal Physiology**

Section 7: **Endocrine Physiology**

Section 1

CELL PHYSIOLOGY, FLUID HOMEOSTASIS, AND MEMBRANE TRANSPORT

***Physiology** is the study of how the systems of the body work, not only on an individual basis, but also in concert to support the entire organism. Medicine is the application of physiologic principles, and understanding these principles gives us insight into the development of disease. The terms* regulation *and* integration *will keep surfacing as you learn more about how each system functions. Because of these building interactions, the field of physiology is always expanding. As we discover more about the genes, molecules, and proteins that regulate other factors, we see that the discipline of physiology is far from static. Each new discovery gives us more insight into how our impossibly complex organism exists, and how we might intercede when pathophysiology occurs. This text will explore essential elements in each of the body's systems; it is not intended to be comprehensive, but focuses, rather, on ensuring a solid understanding of these principles related to the regulation and integration of the systems.*

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Chapter 1

The Cell and Fluid Homeostasis

CELL STRUCTURE AND ORGANIZATION

Organisms evolved from single cells floating in the primordial sea (Fig. 1.1). A key to appreciating how multicellular organisms exist is through understanding how the single cells maintained their internal fluid environment when exposed directly to the outside environment with the only barrier being a semipermeable membrane. Nutrients from the "sea" entered the cell, diffusing down their concentration gradients through channels or pores, and waste was transported out through exocytosis. In this simple system, if the external environment changed (e.g., if salinity increased due to excess heat and evaporation of sea water or water temperature changed), the cell adapted or perished. To evolve to multicellular organisms, cells developed additional barriers to the outside environment to allow better regulation of the intracellular environment.

In multicellular organisms, cells undergo differentiation, developing discrete intracellular proteins, metabolic systems, and products. The cells with similar properties aggregate and become tissues and organ systems [cells → tissues → organs → systems].

Various tissues serve to support and produce movement (muscle tissue), initiate and conduct electrical impulses (nervous tissue), secrete and absorb substances (epithelial tissue), and join other cells together (connective tissue). These tissues combine and support organ systems that control other cells (nervous and endocrine systems), provide nutrient input and continual excretion of waste (respiratory and gastrointestinal systems), circulate the nutrients (cardiovascular system), filter and monitor fluid and electrolyte needs and rid the body of waste (renal system), provide structural support (skeletal system), and provide a barrier to protect the whole structure (integumentary system [skin]) (Fig. 1.2).

THE CELL MEMBRANE

The human body is composed of eukaryotic cells (those that have a true nucleus) containing various organelles (mitochondria, smooth and rough endoplasmic reticulum, Golgi apparatus, etc.) that perform specific functions. The nucleus and organelles are surrounded by a **plasma membrane** consisting of a lipid bilayer primarily made of phospholipids, with varying amounts of glycolipids, cholesterol, and proteins. The lipid bilayer is positioned with the hydrophobic fatty acid tails of phospholipids oriented toward the middle of the membrane, and the hydrophilic polar head groups oriented toward the extracellular or intracellular space. The fluidity of the membrane is maintained in large part by the amount of short-chain and unsaturated fatty acids incorporated within the phospholipids; incorporation of cholesterol into the lipid bilayer reduces fluidity (Fig. 1.3). The oily, hydrophobic interior region makes the bilayer an effective barrier to fluid (on either side), with permeability only to some small hydrophobic solutes, such as ethanol, that can diffuse through the lipids.

To accommodate multiple cellular functions, the membranes are actually **semipermeable** because of a variety of proteins inserted in the lipid bilayer. These proteins are in the form of ion channels, ligand receptors, adhesion molecules, and cell recognition markers. Transport across the membrane can involve passive or active mechanisms and is dictated by the membrane composition, concentration gradient of the solute, and availability of transport proteins (see Chapter 2). If the integrity of the membrane is disrupted by changing fluidity, protein concentration, or thickness, transport processes will be impaired.

FLUID COMPARTMENTS: SIZE AND CONSTITUTIVE ELEMENTS

Fluid Compartments and Size

The typical adult body is approximately 60% water; in a 70-kilogram (kg) person, this equals 42 liters (L) (Fig. 1.4). The actual size of all fluid compartments is dependent on a variety of factors including size and body mass index. In the normal 70-kg adult:

- **Intracellular fluid** (**ICF**) constitutes $^{2}/_{3}$ of the **total body water** (28 L), and the **extracellular fluid** (**ECF**) accounts for the other $^{1}/_{3}$ of total body water (14 L).
- The **extracellular fluid** compartment is composed of the *plasma* (blood without red blood cells) and the **interstitial**

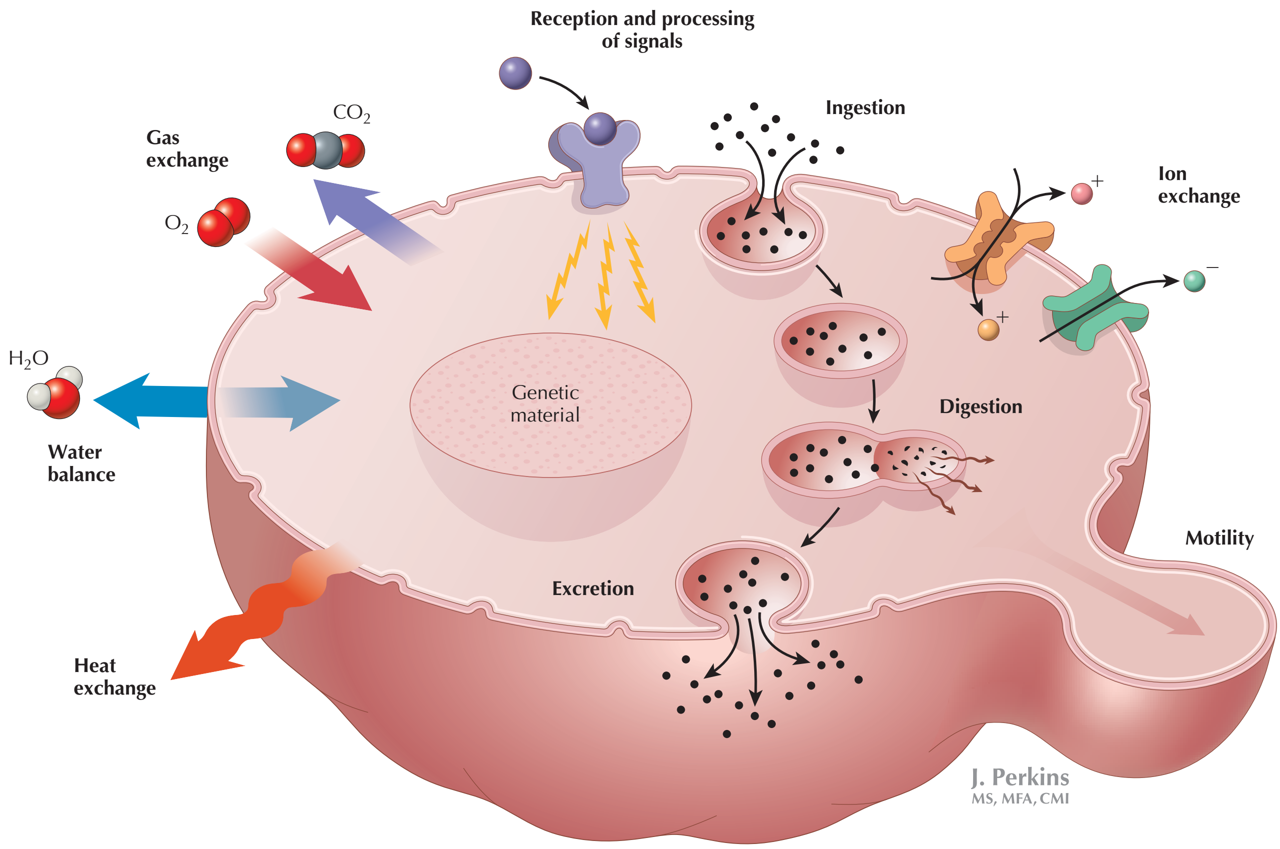


Figure 1.1 **Cell in the Primordial Sea** The first single-celled organisms had to perform basic functions and be able to adapt to changes in their immediate external environment. The semipermeable cell membrane facilitated the processes that provided nutrients to the cell, using diffusion, endocytosis and exocytosis, and protein transporters to maintain homeostasis.

fluid (**ISF**), which is the fluid bathing cells (outside of the vascular system) as well as the fluid in bone and connective tissue. Plasma constitutes $^{1}/_{4}$ of ECF (3.5 L), and ISF constitutes the other $^{3}/_{4}$ of ECF (10.5 L).

The amount of **total body water (TBW)** differs with age and general body type. TBW in rapidly growing infants is ~75% of body weight, whereas older adults have a lower percentage. In addition, body fat plays a role: obese individuals have lower TBW than age-matched individuals, and, in general, women have less TBW than age-matched men. This is especially relevant for drug dosages. Because fat solubility varies with the type of drug, body water composition (relative to body fat) can affect the effective concentration of the drug (Fig. 1.5).

Intracellular and Extracellular Compartments

The intracellular and extracellular compartments are separated by the cell membrane. Within the ECF, the plasma and interstitial fluid are separated by the endothelium and basement membranes of the capillaries. The ISF surrounds the cells and is in close contact with both the cells and the plasma.

The ICF has different solute concentrations than the ECF, primarily due to the Na^+ pump, which maintains an ECF high in Na^+, and an ICF high in K^+ (Fig. 1.6). The maintenance of different solute concentrations is also highly dependent on the selective permeability of cell membranes separating the extracellular and intracellular spaces. The cations and anions in our body are in balance, with the number of positive charges in each compartment equaling the number of negative charges (see Fig. 1.6). Because the ion flow across the membrane is responsive to both the electrical charge and the solute gradient, the overall environment is controlled by maintenance of this **electrochemical equilibrium.**

The osmolarity (total concentration of solutes) of fluids in our bodies is ~290 milliosmoles (mosm)/L (generally rounded to 300 mosm/L for calculations). This is true for all of the fluid compartments (see Fig. 1.6). The basolateral sodium ATPase pumps (seen on cell membranes) are instrumental in establishing and maintaining the intracellular and extracellular

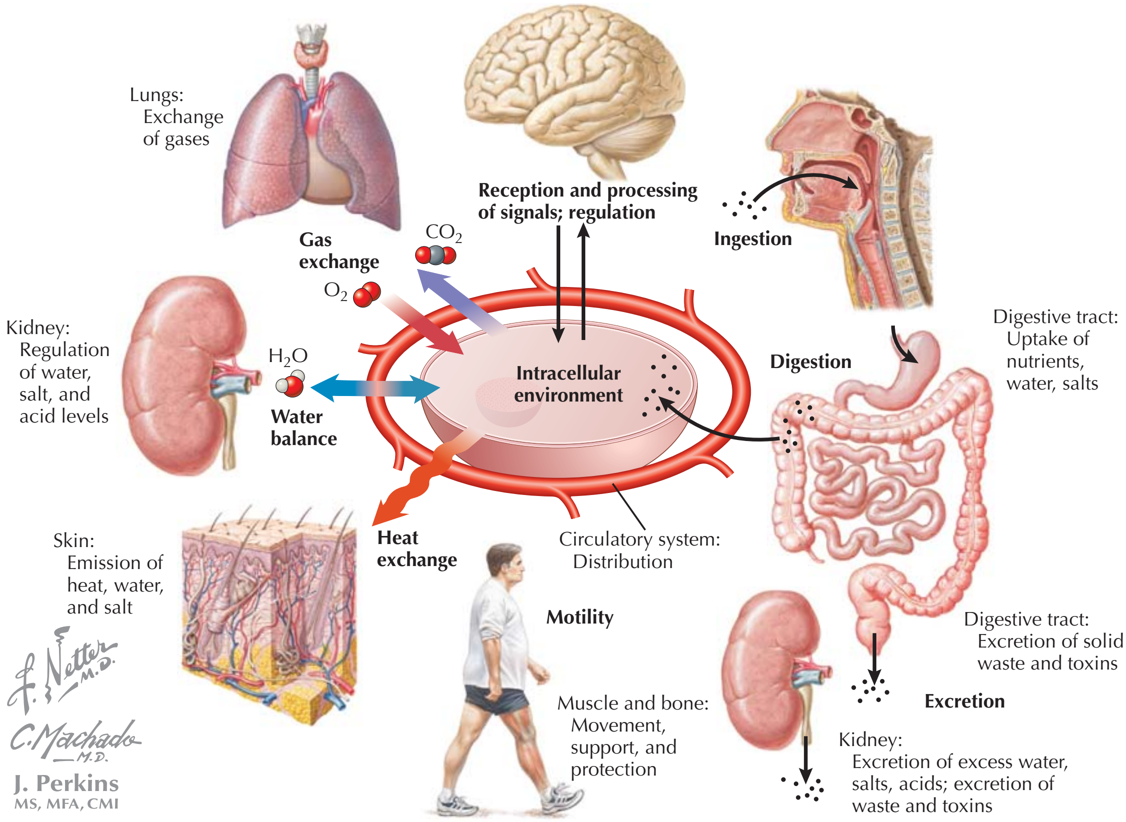


Figure 1.2 **Buffering the External Environment** In multicellular organisms, the basic homeostatic mechanisms of single-celled organisms are mirrored by integration of specialized organ systems to create a stable environment for the cells. This allows specialization of cellular functions and a layer of protection for the systems.

environments. Intracellular Na^+ is maintained at a low concentration (which drives the Na^+-dependent transport into the cells) compared with the high Na^+ in ECF. The **extracellular sodium** (and the small amount of other positive ions) is balanced by chloride and bicarbonate anions and anionic proteins. For the most part, the concentration of solutes between plasma and ISF is similar, with the exception of proteins (indicated as A^-), which remain in the vascular space (under normal conditions, they cannot pass through the capillary membranes). The high ECF Na^+ concentration drives Na^+ leakage into cells, as well as many other transport processes.

The primary intracellular cation is *potassium ion,* which is balanced by phosphates, proteins, and small amounts of other miscellaneous anions. Because of the high concentration gradients for sodium, potassium, and chloride, there is passive leakage of these ions down their gradients. The leakage of potassium out of the cell through specific K^+ channels is the key factor contributing to the resting membrane potential. The differential sodium, potassium, and chloride concentrations across the cell membrane are crucial for the generation of electrical potentials (see Chapter 3).

OSMOSIS, STARLING FORCES, AND FLUID HOMEOSTASIS

Osmosis

Membranes are **selectively permeable (semipermeable),** meaning they allow some, but not all, molecules to pass through. Membranes of tissues vary in their permeability to specific solutes. This tissue specificity is critical to function, as seen in the variation in cell solute permeability through a renal nephron (see Chapters 17 and 18). On either side of the membrane, there are factors that oppose and facilitate movement of water and solutes out of the compartments. These factors include:

- Concentration of specific solutes. Higher concentration of a solute on one side of the membrane will favor movement of that solute to the other by diffusion.
- Overall concentration of solutes. Higher osmolarity on one side provides osmotic pressure “pulling” water into that space (diffusion of water).
- Concentration of proteins. Because the membrane is impermeable to proteins, protein concentration

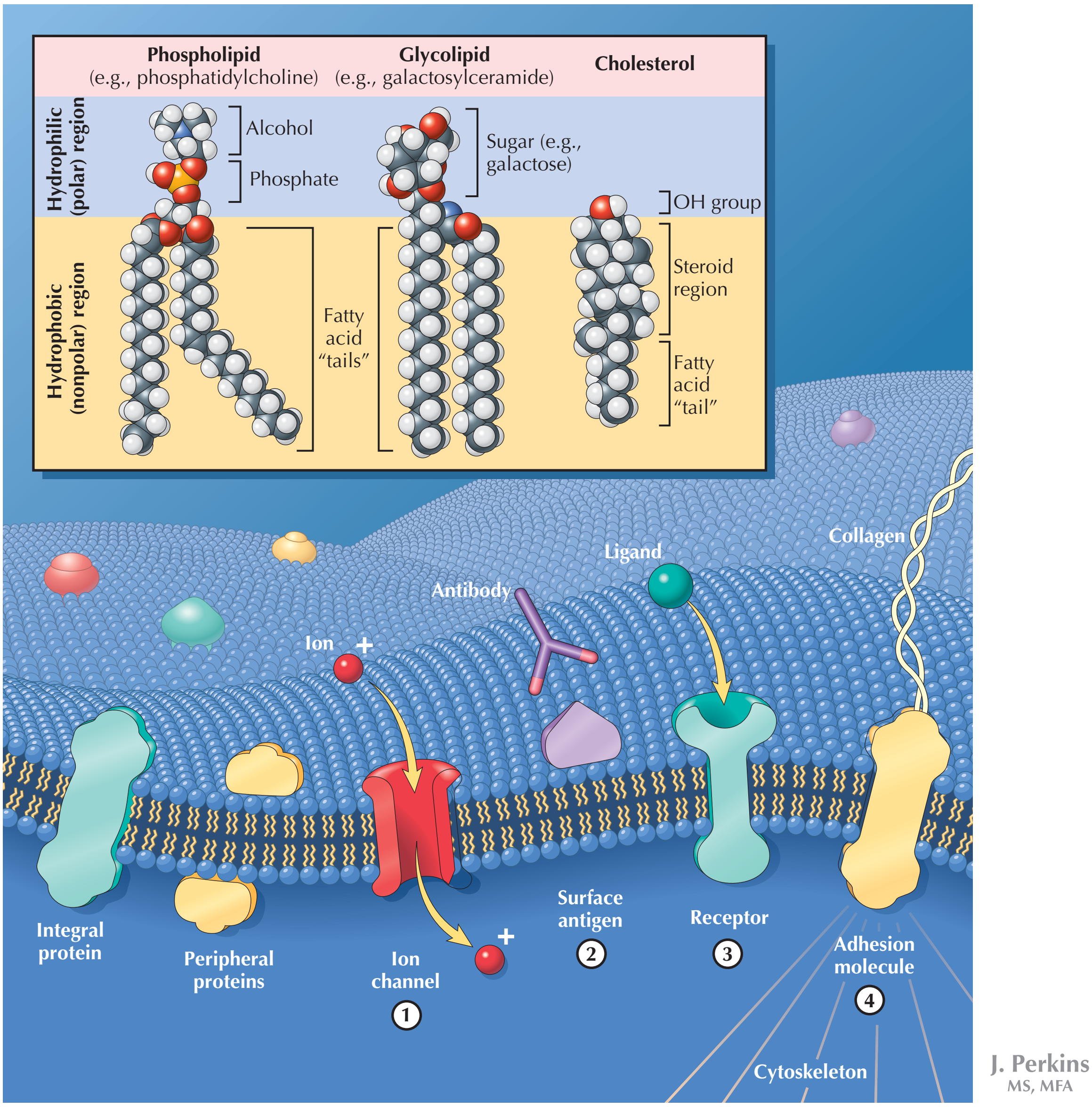


Figure 1.3 **The Eukaryotic Plasma Membrane** The plasma membrane is a lipid bilayer, with hydrophobic ends oriented inward and hydrophilic ends oriented outward. Primary constituents of the membrane are phospholipids, glycolipids, and cholesterol. There are a wide variety of proteins associated with the membrane, including *(1)* ion channels, *(2)* surface antigens, *(3)* receptors, and *(4)* adhesion molecules.

establishes an oncotic pressure "pulling" water into the space with higher concentration.

- Hydrostatic pressure, which is the force "pushing" water out of the space, for example, from capillaries to ISF (when capillary hydrostatic pressure exceeds ISF hydrostatic pressure).

If the membrane is permeable to a solute, **diffusion** of the solute will occur down the concentration gradient (see Chapter 2). However, if the membrane is not permeable to the solute, the solvent (in this case water) will be "pulled" across the membrane toward the compartment with higher solute concentration, until the concentration reaches equilibrium across the membrane. The movement of water across the membrane by diffusion is termed **osmosis,** and the permeability of the membrane determines whether diffusion of solute or osmosis (water movement) occurs. The concentration of the impermeable solute will determine how much water will move through the membrane to achieve **osmolar equilibration** between ECF and ICF.

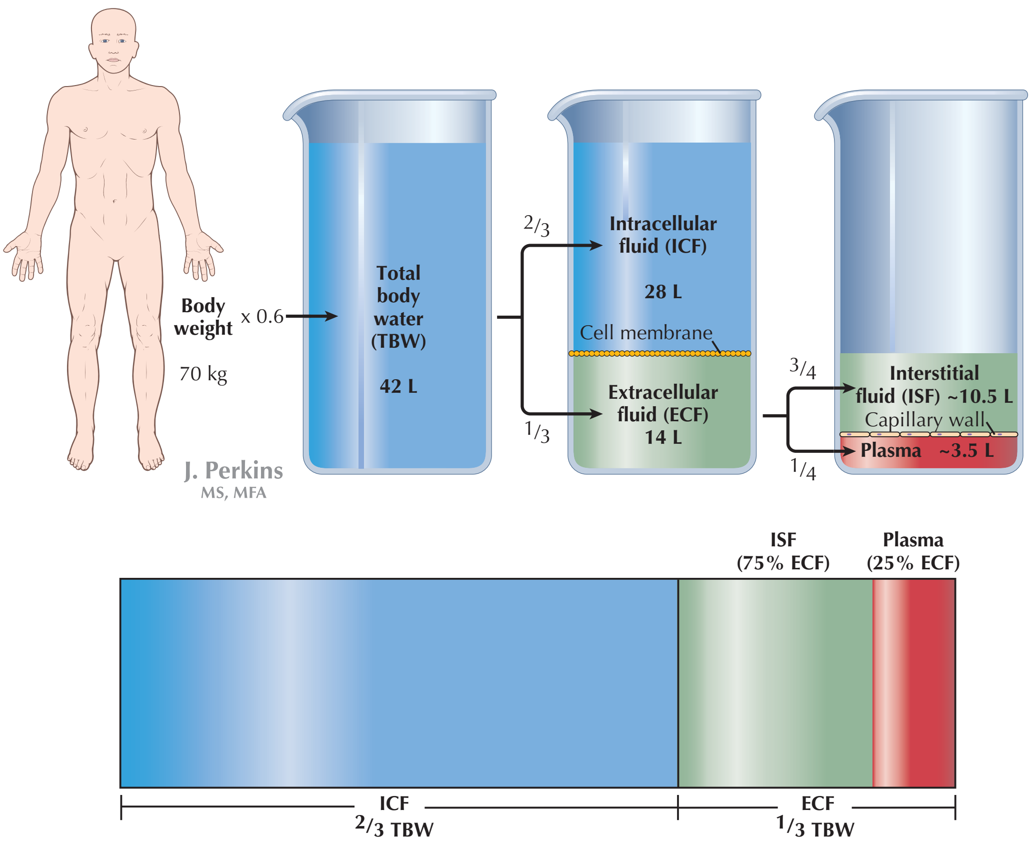


Figure 1.4 Body Fluid Compartments Under normal conditions the total volume of water in the human body (TBW) is about 60% of the body weight. Of TBW, most ($^2/_3$) is intracellular fluid (ICF), and $^1/_3$ is extracellular fluid (ECF). The extracellular fluid is made up of plasma and interstitial fluid (ISF).

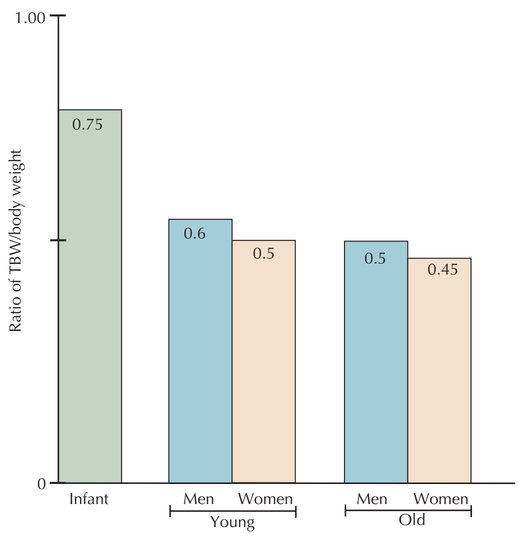


Figure 1.5 Total Body Water as Function of Body Weight Under normal conditions, total body water is most affected by the amount of body fat, and there is more body water as a percentage of body weight in infants and women (because of estrogens). Aging also decreases the ratio because of reduced muscle mass.

Osmosis occurs when **osmotic pressure** is present. This is equivalent to the hydrostatic pressure necessary to prevent movement of fluid through a semipermeable membrane by osmosis. The idea can be illustrated using a U-shaped tube with different concentrations of solute on either side of an **ideal semipermeable membrane** (where the membrane is permeable to water but is impermeable to solute) (Fig. 1.7*A*).

Because of the unequal solute concentrations, fluid will move to the side with the higher solute concentration (right side of tube), against the gravitational force (hydrostatic pressure) that opposes it, until the **hydrostatic pressure** generated is equal to the **osmotic pressure.** In the example, at equilibrium, solute concentration is nearly equal and water level is unequal, and the displacement of water is due to osmotic pressure (Fig. 1.7*B*).

In the plasma, the presence of proteins also produces a significant **oncotic pressure,** which opposes hydrostatic pressure (filtration out of the compartment) and is considered the **effective osmotic pressure of the capillary.**

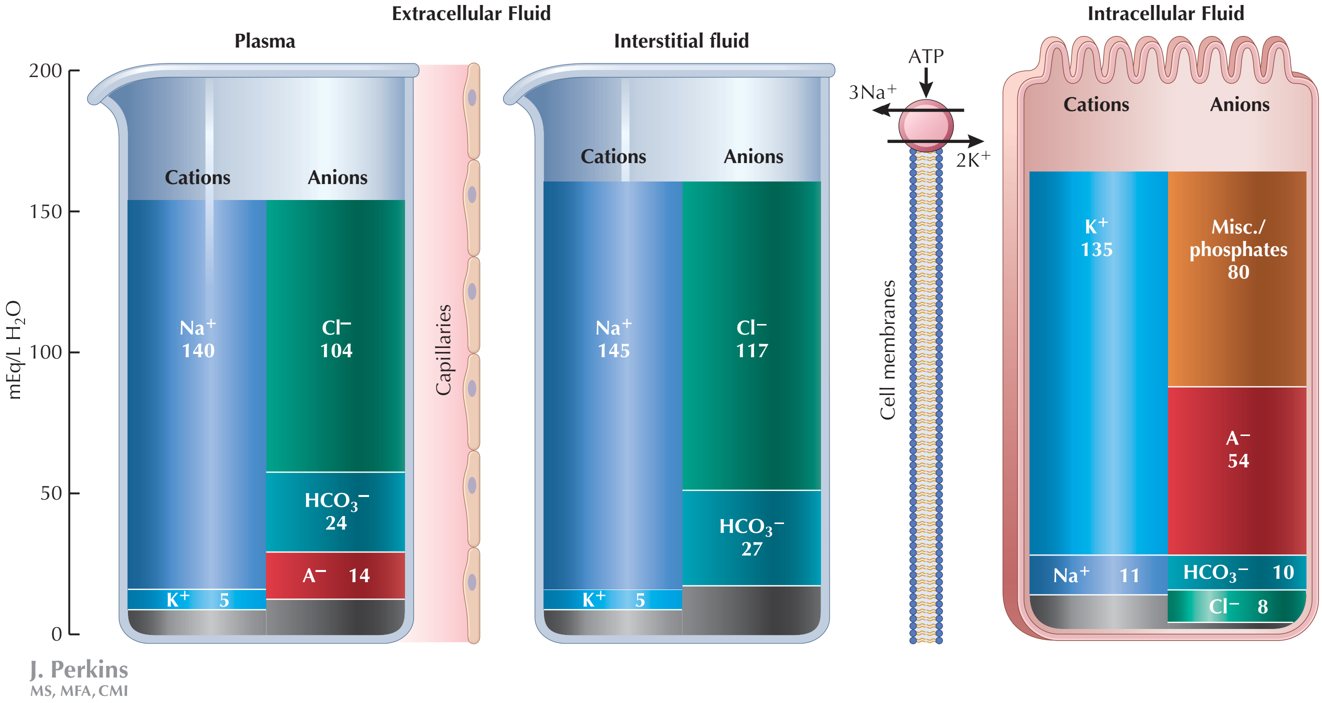


Figure 1.6 **Electrolyte Concentration in Extracellular and Intracellular Fluid** The primary extracellular fluid (ECF) cation is sodium, and the primary interstitial fluid (ICF) cation is potassium. This difference is maintained by the basolateral Na^+/K^+ ATPases, which transport three Na^+ molecules out of the cell in exchange for two K^+ molecules transported into the cell. A balance of positive and negative charges is maintained in each compartment, but by different ions. (Values are approximate.)

Starling Forces

The oncotic and hydrostatic pressures are key components of the **Starling forces.** Starling forces are the pressures that control fluid movement across the capillary wall. Net movement of water out of the capillaries is **filtration,** and net movement into the capillaries is **absorption.** As seen in Figure 1.8, there are four forces controlling fluid movement:

- HP_c, the **capillary hydrostatic pressure,** favors movement out of the capillaries and is dependent on both arterial and venous blood pressures (generated by the heart).
- π_c, the **capillary oncotic pressure,** opposes filtration out of the capillaries and is dependent on the protein concentration in the blood. The only effective oncotic agent in capillaries is protein, which is ordinarily impermeable across the vascular wall.
- P_i, the **interstitial hydrostatic pressure,** opposes filtration out of capillaries, but normally this pressure is low.
- π_i, the **interstitial oncotic pressure,** favors movement out of the capillaries, but under normal conditions, there is little loss of protein out of the capillaries, and this value is near zero.

Movement of fluid through capillary beds can differ due to physical factors particular to the capillary wall (e.g., pore size, fenestration) and its relative permeability to protein, but in general these factors are considered constant for most tissues.

These forces are used to describe net filtration using the ***Starling Equation***,

$$\text{Net filtration} = K_f\left[(HP_c - P_i) - \sigma(\pi_c - \pi_i)\right]$$

in which the constant, K_f, accounts for the physical factors affecting permeability of the capillary wall, and **σ** describes the permeability of the membrane to proteins (where $0 < \sigma < 1$). The liver capillaries (sinusoids) are highly permeable to proteins, and $\sigma = 0$. Thus, bulk movement in the liver sinusoids is controlled by hydrostatic pressure. In contrast, capillaries in most tissues have low permeability to proteins, and $\sigma = {\sim}1$, so the Starling equation can easily be viewed as the pressures governing filtration minus those favoring absorption:

$$(\text{Filtration}) - (\text{Absorption})$$

$$\textbf{Net Filtration} = K_f\left[(HP_c + \pi_i) - (P_i + \pi_c)\right]$$

Although K_f is a "constant," it differs between systemic, cerebral, and renal glomerular capillaries, with cerebral

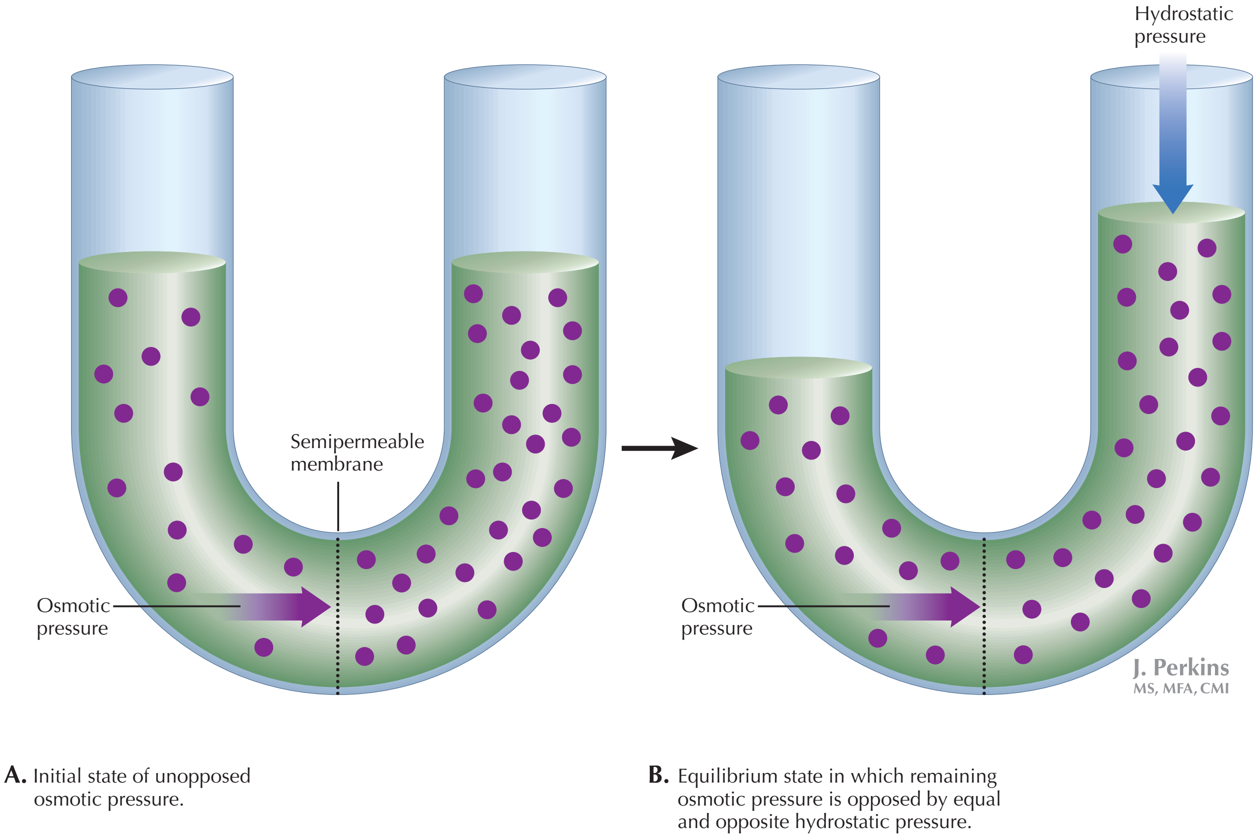


Figure 1.7 **Osmosis and Osmotic Pressure** When a semipermeable membrane separates two compartments in a "V tube," fluid will move through the membrane toward the higher solute concentration *(A),* until near equilibrium of solutes is reached and the remaining osmotic pressure is opposed by the hydrostatic pressure difference between the two sides of the tube *(B).* In blood vessels, hydrostatic pressure is generated by gravity and the pumping of the heart, and the osmotic pressure is measured as the force needed to oppose the hydrostatic pressure. Osmotic pressure works toward equalization of solute concentrations on either side of the membrane. Oncotic pressure (π) is the osmotic pressure produced by impermeable proteins. In the plasma, oncotic pressure is considered the effective osmotic pressure of the capillary.

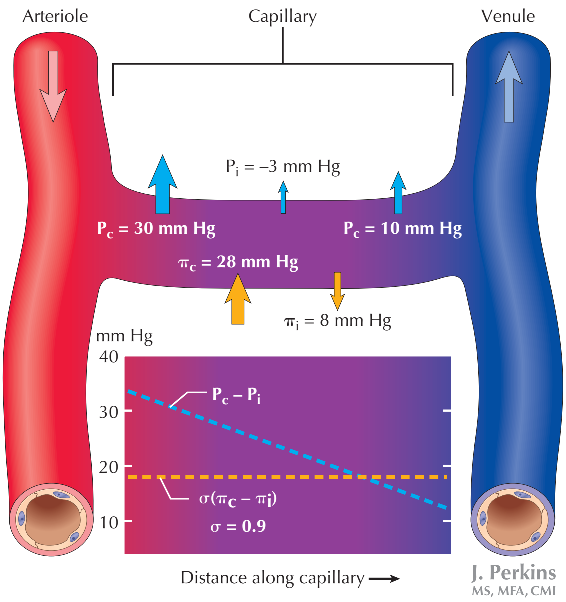
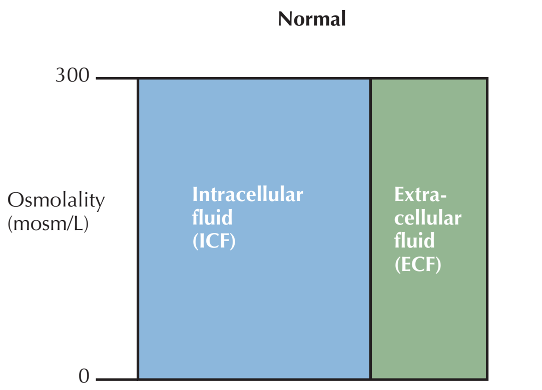
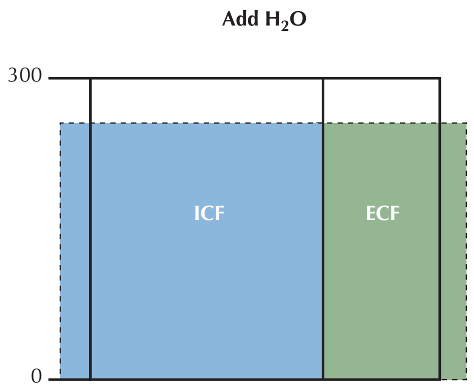


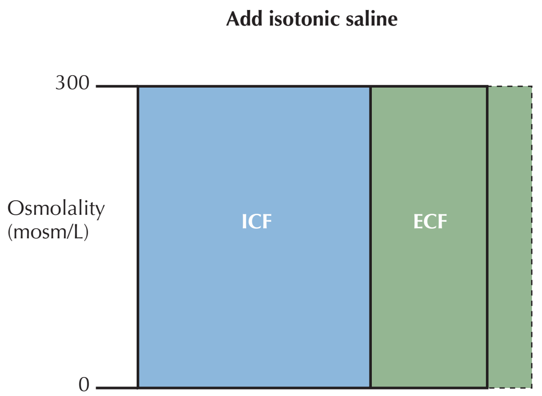
Figure 1.8 **Starling Forces across the Capillary** The Starling forces (hydrostatic and oncotic pressures) allow for bulk flow of fluid and nutrients across the capillary wall. The permeability of the membranes of the capillary wall to proteins is usually very low in most tissues and is reflected by a protein reflection coefficient (σ) of ~1. The inset panel illustrates that as fluid moves though the capillaries and diffusion into tissues occurs, the Starling forces change, and the forces favoring net filtration (especially P_c [HP_c, capillary hydrostatic pressure]) decrease *(dotted blue line).*



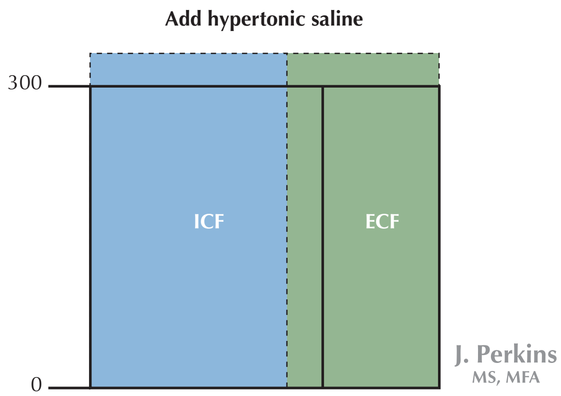
A. Under normal conditions, ICF is 2/3 and ECF is 1/3 of TBW. The osmolality of both compartments is ~300 mosm/L.



B. If **water** was added to the plasma (thus, ECF), ECF osmolality would be diluted initially (become **hypotonic**) compared to ICF. Water would then enter the ICF space (cells would swell) to equilibrate the osmolality between compartments. The overall effect would be to reduce the osmolality of ICF and ECF and expand the compartment size.



C. If **isotonic saline** (solution with NaCl concentration of ~300 mosm/L) was added to the plasma (ECF), the fluid will stay in the ECF because it is isotonic, expanding that compartment.



D. If **hypertonic saline** was added to the plasma, ECF osmolality would increase greatly initially and fluid would be drawn out of the cells and into the ECF to lower the tonicity of the ECF. This would contract the volume of the ICF compartment (cells would shrink) and increase the volume of the ECF compartment, as well as increase overall osmolality.

Figure 1.9 Effect of Adding Solutes to the Extracellular Fluid on Compartment Size If there were not compensatory mechanisms in place to sense and regulate plasma volume and osmolarity, addition of water and hypertonic solutions would have a profound effect on ICF volume and tonicity. *A* illustrates the intracellular fluid (ICF) and extracellular fluid (ECF) compartment sizes under normal osmolar conditions (300 mosm/L). *B* through *D* illustrate the effect of changing the tonicity (the osmolarity relative to plasma) of the extracellular fluid on cells. *B,* Addition of pure water to the ECF will increase both ECF and ICF volume, and reduce overall osmolarity. *C,* Addition of isotonic (same osmolarity as plasma) saline (NaCl) to the ECF will expand only the ECF compartment size, because NaCl is mainly ECF electrolytes. *D,* Addition of pure NaCl will increase overall osmolarity of both compartments to a new level. ICF will shrink, as water is drawn into the ECF toward the higher osmolar concentration, and the volume of the ECF will expand by addition of the ICF fluid.

capillaries having a lower K_f (limiting filtration), and glomerular capillaries having a greater K_f (promoting filtration) compared with systemic capillaries. Thus, filtration will be determined by the difference in hydrostatic pressure between the capillary and interstitium, minus the difference between capillary and interstitial oncotic pressure (corrected for the protein reflection coefficient). It should be clear that under normal conditions, the forces that are most variable are the HP_c and the π_c, because those can reflect changes in plasma volume.

Examples of the effects of Starling forces on fluid shifts can be illustrated by changes in fluid volume as well as changes in physical factors. A severely dehydrated person will have a decreased blood volume, which would potentially lower blood pressure (e.g., HP_c) and increase π_c. Looking at the Starling equation, those changes will decrease filtration force and increase the absorptive force, causing an overall decrease in net filtration. This will keep fluid in the vascular space.

When physical attributes of the membrane change, Starling forces are also affected: K_f can change if the capillary membrane is damaged, such as by toxins or disease. If the clefts between endothelial cells or fenestrations expand (as seen in diseased renal glomeruli), plasma proteins may pass into the interstitial space and alter the Starling forces by increasing π_i: In peripheral capillaries, this causes **edema.** Starling forces are also affected in congestive heart failure (CHF), cirrhosis, and sepsis.

Homeostasis

French physiologist Claude Bernard first articulated the concept that maintaining a constant internal environment, or **milieu intérieur,** was essential to good health. In multicellular organisms, the balance between internal and external environments is critical, and the ability to maintain a constant internal function during changes in the external environment is termed **homeostasis.** This is accomplished through integrated regulation of the internal environment by the multiple organ systems (see Fig. 1.2).

On the cellular level, homeostasis is possible due to expandable semipermeable membranes, which can accommodate small changes in osmolarity via osmosis. However, for proper cellular function, the intracellular fluid, and thus osmolarity, must be kept under tight control.

The plasma is the interface between the internal and the external environment; therefore, maintaining plasma osmolarity is an important key to cell homeostasis. Because of this, many systems play a role in controlling plasma osmolarity. Both **thirst** and the **salt appetite** are behavioral responses that can be stimulated by dehydration and/or blood loss. These serve to stimulate specific ingestive behaviors (drinking, or eating salty foods that will also stimulate drinking) that will increase the input of fluid and salt to the system. On a minute-to-minute basis, the endocrine and sympathetic nervous systems work to regulate the amount of sodium and water retained by the kidneys, thus controlling plasma osmolarity (Fig. 1.9).

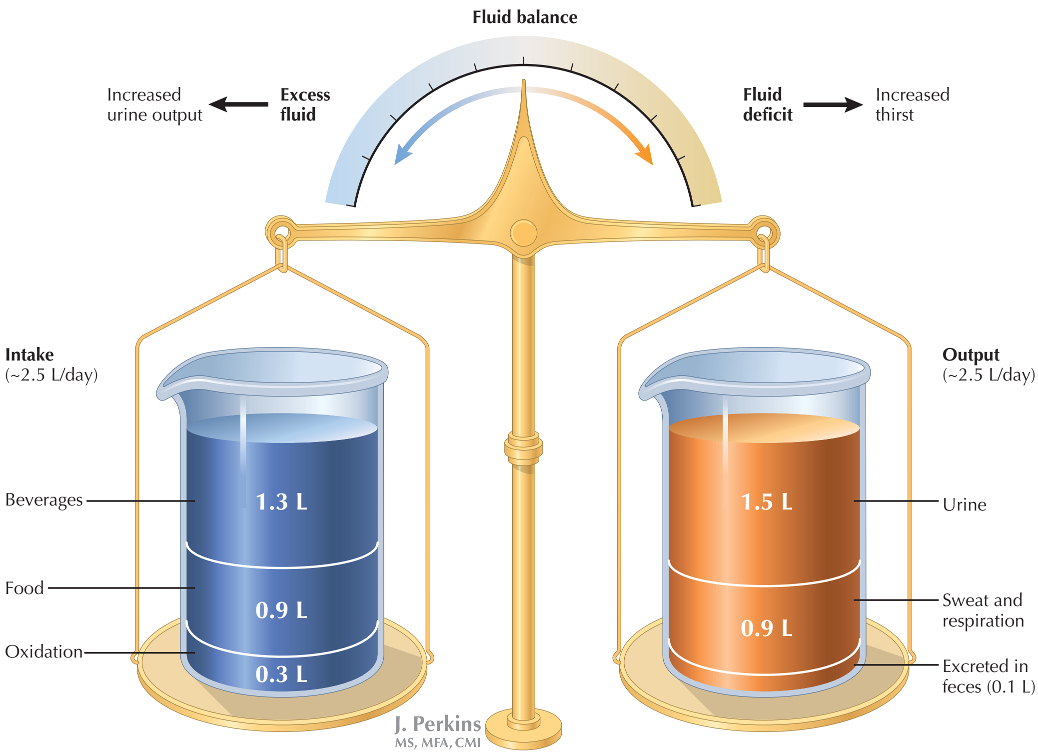


Figure 1.10 Net Fluid Balance To maintain fluid balance, fluid input needs to match fluid output. If intake (from food and beverages) exceeds output (urine, fluid in feces, insensible losses), the organism is in positive balance, and urine volume will increase to eliminate excess fluid. Negative fluid balance occurs when intake is less than output. In this case, integrated responses will increase thirst and decrease additional fluid losses, until homeostasis is restored.

Normally, changes in plasma osmolarity are well controlled and homeostasis is maintained as a result of hypothalamic osmoreceptors and the kidneys sensing fluid composition; carotid and aortic baroreceptors sensing pressure; release of hormones in response to altered pressure and osmolarity; and the actions of the kidney to regulate sodium and water reabsorption. This integrated control is the key to fluid homeostasis. Control of renal fluid and electrolyte handling will be covered in Section 5.

Fluid intake and output must be in balance (Fig. 1.10). If water intake (through food and fluids) is greater than the output (urine and insensible losses from sweat, breathing, and feces), the organism has a surplus of fluid, which will decrease plasma osmolarity, and the kidneys will excrete the excess fluid (see Section 5). Conversely, if the intake is less than output, the organism has a deficit of fluid, and plasma osmolarity will increase. In this situation, the thirst response will be activated and the kidneys will conserve fluid, producing less urine. This idea of balance is expanded on in the following sections, and the integration of the endocrine, cardiovascular, and renal systems in regulation of fluid and electrolyte homeostasis is discussed more fully.

CLINICAL CORRELATE
Measurement of Fluid Compartment Size

The indicator-dilution method is used to determine the volume of fluid in the different fluid compartments. This is done using indicators specific to each compartment. A known quantity of the substance is infused into the bloodstream of the subject and allowed to disperse. A plasma sample is then obtained, and the amount of indicator determined. The compartment volume is then calculated by the formula:

$$\text{Volume (in liters)} = \frac{\text{amount of indicator injected (mg)}}{\text{final concentration of indicator (mg/L)}}$$

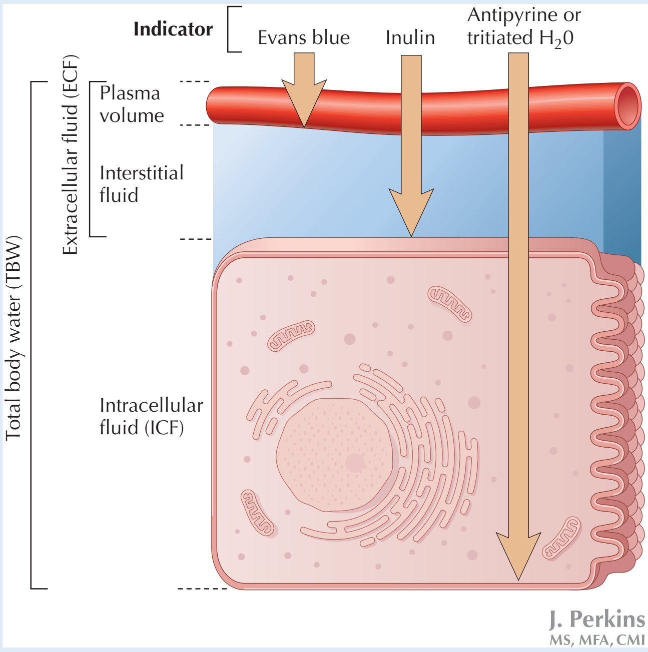
Compartment	Indicator
TBW	Antipyrine or tritiated water, because both of these substances will diffuse through *all* compartments.
ECF	Inulin, which will diffuse throughout plasma and ISF. Inulin is a large sugar (MW 500) that cannot cross cell membranes and is not metabolized.
Plasma volume	Evans blue dye, which binds to plasma proteins. The total blood volume is comprised of the plasma and red blood cells, and the hematocrit is the percentage of red blood cells (RBC) in the whole blood. Hematocrit is ~0.42 (42% RBC) in normal adult men, and ~0.38 in women.

By extrapolation, the other compartments can be determined:

$$\text{Since TBW} = \text{ECF} + \text{ICF},$$
$$ICF = \text{TBW} - \text{ECF}$$
$$ISF = (\text{ECF} - \text{Plasma Volume})$$

Because blood volume = plasma volume + red blood cell volume (see earlier), it can be calculated by the formula:

$$\textit{Blood Volume} = (\text{Plasma Volume} \div [1 - \text{hematocrit}])$$



Substances Used to Determine Fluid Compartment Size

Chapter 2 Membrane Transport

CELLULAR TRANSPORT: PASSIVE AND ACTIVE MECHANISMS

Ions and solutes move through several different types of carrier proteins and channels that allow solute movement through the plasma membrane in several different ways. The carriers and channels include:

- **Ion channels** and **pores** that allow diffusion of solutes between compartments.
- **Uniporters,** which are membrane transport proteins that recognize specific molecules, such as fructose.
- **Symporters** that transport a cation (or cations) down its concentration gradient with another molecule (either another ion or a sugar, amino acid, or oligopeptide).
- **Antiporters** that transport an ion down its concentration gradient while another substance is transported in the opposite direction. This type of transport is often associated with Na^+ transport, or may be dependent on gradients of other ions, as in the case of the HCO_3^-/Cl^- exchanger.

Transporters (or carrier proteins) can be energy dependent or independent. Movement through channels and uniporters follows the concentration gradient of the molecule or the electrochemical gradient established by movement of other ions. However, most movement in nonexcitable cells occurs though some expenditure of energy, either by primary or secondary active transport.

Passive Transport

Regardless of the type of carrier or channel involved, if no energy is expended in the transport process, it is considered **passive transport**. Passive transport can occur via **simple** or **facilitated diffusion.**

Simple Diffusion

If a substance is lipid soluble (a property of gases, some hormones, and cholesterol), it will move down its concentration gradient through the cell membrane by **simple diffusion** (Fig. 2.1). This movement is described by ***Fick's law***.

Fick's Law

$$J_i = D_i \times A(1/X) \times (C_1 - C_2)$$

Where:

- J_i represents net flux
- D_i is the coefficient of diffusion
- A is the area
- X is the distance through the membrane
- $(C_1 - C_2)$ is the difference in concentration across the membrane

Thus, passive diffusion of a molecule across a membrane will be directly proportional to the surface area of the membrane and the difference in concentration of the molecule, and inversely proportional to the thickness of the membrane.

Facilitated Diffusion

Facilitated diffusion can occur through either **gated channels** or carrier proteins in the membrane. Gated channels are pores that have "doors" that can open or close in response to external elements, regulating the flow of the solute (Fig. 2.2*A*). Examples include Ca^{2+}, K^+, and Na^+. This type of gated transport into and out of the cell is critical to most membrane potentials, except the resting potential (see Chapter 3). When facilitated diffusion of a substance involves a carrier protein, binding of the substance to the carrier results in a conformational change in the protein and translocation of the substance to the other side of the membrane.

Simple and facilitated diffusion do not require expenditure of energy, but do depend on the size and composition of the membrane and the concentration gradient for the solute. Key differences between these two types of diffusion include:

- **Simple diffusion** occurs over all concentration ranges at a rate linearly related to the concentration gradient—as the concentration gradient increases, the rate of diffusion from the compartment with high concentration to the compartment with low concentration will increase.
- **Facilitated diffusion** is subject to a maximal rate of uptake (V_{max}). The rate of facilitated diffusion is greater than that of passive diffusion at lower solute concentrations. However, at higher solute concentrations the rate of facilitated uptake reaches its V_{max} (carrier is